Partitioning Around Medoids

2026-04-17

Introduction

Partitioning around medoids (PAM) is the canonical k-medoids clustering algorithm. Like k-means, it partitions n observations into k clusters; unlike k-means, the cluster centres are actual observations (medoids) rather than coordinate means. This single change makes PAM robust to outliers (a single extreme value cannot drag a medoid away from the cluster bulk), enables clustering on arbitrary dissimilarity matrices including those derived from non-Euclidean distances or mixed numeric-categorical data, and produces interpretable centre observations that can be quoted in clinical or biological contexts ("the typical patient in cluster 2 is patient 47").

Prerequisites

A working understanding of k-means, distance and dissimilarity matrices, and the difference between mean-based and median-like cluster centres.

Theory

PAM alternates two steps:

1. **Build / swap**: for each cluster, choose the medoid that minimises the total dissimilarity to all other cluster members.
2. **Assign**: assign each observation to the cluster of its nearest medoid.

The objective is the sum of dissimilarities to the assigned medoid, summed over all observations. The algorithm converges to a local minimum; multiple random initialisations help find a better local optimum. Because PAM consumes a dissimilarity matrix, it works with Gower distance for mixed numeric-categorical data, Manhattan distance for outlier-robust clustering, correlation-based distances for gene-expression patterns, and any custom dissimilarity that the analyst can define.

Assumptions

A meaningful dissimilarity (or distance) on the input space. Approximately equal cluster sizes are preferable; PAM does not handle very imbalanced clusters as gracefully as model-based methods (Gaussian mixtures).

R Implementation

```r
library(cluster)

# Mixed-type data with Gower distance
d_iris <- iris
set.seed(2026)
d_iris$Category <- factor(sample(c("A", "B", "C"), 150, replace = TRUE))

g <- daisy(d_iris, metric = "gower")
pam_fit <- pam(g, k = 3)
```

```r
pam_fit$medoids
table(pam_fit$clustering, iris$Species)

# Silhouette
summary(silhouette(pam_fit))
```

Output & Results

`pam()` returns medoid identifiers (actual case indices), cluster assignments, and within-cluster summaries. `silhouette()` returns per-observation silhouette widths and an average — a primary internal validation metric.

Interpretation

A reporting sentence: "Partitioning around medoids on Gower-distance dissimilarities produced three clusters that closely match the iris species (Cluster 1: setosa; Cluster 2: 92 % versicolor; Cluster 3: 86 % virginica) with average silhouette width 0.45 (range -0.10 to 0.78)." Always pair PAM with a silhouette analysis to verify that the chosen k produces well-separated clusters.

Practical Tips

- Choose PAM over k-means when outliers are likely or when the data are mixed-type or genuinely non-Euclidean; the medoid is robust where the mean is not.
- `cluster::pam()` accepts either a data matrix (computing Euclidean distance internally) or a pre-computed `dist` object via the `diss = TRUE` argument.
- For large datasets ($n > 1{,}000$), `cluster::clara()` runs PAM on sub-samples and combines results, scaling to tens of thousands of observations.
- Use `fpc::pamk()` to automatically select k via the silhouette method; it returns the optimal k along with the fit.
- For Gower distances on mixed numeric and categorical data, `cluster::daisy(metric = "gower")` is the standard input; categorical variables become indicator distances, numeric ones become Manhattan after range normalisation.
- For very large n, alternatives such as CLARANS, BIRCH, or mini-batch k-means scale better; the price is reduced robustness or interpretability.

R Packages Used

`cluster` for `pam()`, `clara()`, `daisy()`, and `silhouette()`; `fpc` for `pamk()` automatic k selection; `factoextra` for `fviz_cluster()` ggplot-style cluster visualisation; `klaR::kmodes()` for an analogous categorical-only k-modes algorithm.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based

- UMAP and t-SNE